

GENERAL SPECIFICATION G6.20

Data Communications Networks (for TWRP Facilities)

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G6.20 DATA COMMUNICATIONS NETWORKS (FOR TWRP FACILITIES)

G6.20.1 General

- (a) Three fibre optic data communications backbones will link all major areas of the plant. These shall be used to form the following separate networks:
 - (i) Process Monitoring Network (PMN) and Process Automation Networks (PAN);
 - (ii) Non-Process Networks; and
 - (iii) Fire protection network.
- (b) Where specified in the Particular Specifications or Drawings, the Contractor shall provide networks shall within each plant area as follows:
 - (i) PAN sub-ring networks connecting PLC's across multiple XLV rooms; and
 - (ii) Field networks to connect PLC's with remote i/o, machine control panels and MCC's. Some areas shall have multiple field networks.
- (c) Process, Non-Process, and Fire Networks shall use separate fibre cables to allow fibre optic breakout trays (FOBOTs) and patch panels to be segregated.
- (d) Cables for process and non-process networks may share common conduits.
- (e) Cable for redundant network pairs shall not share common conduits.
- (f) Ethernet-based networks shall be used for all backbone networks, sub-rings, and all field networks.
- (g) Networks shall be capable of communication speeds shown on the Drawings or in Particular Specifications.
- (h) For Ethernet based networks, peak loading shall not exceed 30% of maximum capacity so as to ensure the network is effectively deterministic.
- (i) All connections exceeding 100 m in length shall be by optic fibre, as shall be any connections depicted in Drawings as being by optical fibre.
- (j) The Contractor shall allow for any limitation in distance imposed by various media and protocols and shall design the data communication networks accordingly.
- (k) The Contractor shall provide all necessary components for the complete functioning of networks including but not limited to FOBOTS, fibre optic transceivers, fibre optic to copper converters.
- (l) Wherever the Contractor provides switches, servers, workstations, HMI's and SCADA Applications, they shall configure these in accordance with the Board's ICS Security Implementation Guide. A copy will be provided after Award if required.

G6.20.2 Process Monitoring Network and Process Automation Networks

- (a) PAN sub-rings shall be cabled as 2 x 24c multi-mode fibre.

G6.20.3 Field Networks

- (a) Field-networks for i/o shall use a fault-tolerant ring architecture.
- (b) PLC I/O shall have one communication path onto the field network.
- (c) Field-networks shall incorporate MCC's and VFD's.
- (d) Field-networks may incorporate some PLC's for complex machines.
- (e) The Contractor shall develop a final design for field networks taking into account the network loading and protocol overhead imposed by the specific equipment chosen.
- (f) The Contractor shall provide network calculations to prove the feasibility of their design.

G6.20.4 Non-Process Networks

- (a) The non-process network backbone shall use a fault-tolerant ring architecture at the back-bone level.
- (b) Below the back-bone level, switches at the network edge, and connected devices require neither redundant nor fault-tolerant connections.
- (c) Refer to General Specification G7 Building Services for additional requirements.

G6.20.5 Fire Protection Network

- (a) Fire networks shall use fire-rated cable and separate FIBOT enclosures meeting fire system requirements.
- (b) Where permissible under fire equipment standards, fire-systems shall communicate using a fault-tolerant ring architecture at the back-bone level.
- (c) Refer to General Specification G7 Building Services for detailed requirements.

G6.20.6 Cable Route Redundancy

- (a) The cables for the redundant ring of each network shall not use the same cable duct (pipe). The Contractor shall provide separate cables for each side of each redundant network. Each cable shall have 100% spare (unused) fibres/cores and there shall be no point at which both sides of a redundant network occupy the same duct or tray. If direct buried but not concrete encased, each side of a redundant network shall be spaced by at least 1 metre in the horizontal plane for the other side of the same network.

G6.20.7 Network Monitoring Systems

- (a) Network Monitoring System (NMS) software shall allow remote configuration, management, fault detection and location and performance monitoring of all communication devices on the network. The NMS shall support all network topologies including linear, star and ring.

- (b) Network monitoring system software shall provide the following:
 - (i) Automated status monitoring of all switches including the collection of alarm events from each item of network equipment. The alarm messages shall be displayed on the NMS user interface as well as be configured to send out as an SNMP trap or email according to the severity of the alarm or warning;
 - (ii) Monitoring of end-point devices;
 - (iii) Automated transmission of alarms by email;
 - (iv) Secure remote management of all switches;
 - (v) Scheduled automatic backup of switch configurations;
 - (vi) Remote downloading of switch configurations;
 - (vii) Both automatic and manual plotting of the network topology;
 - (viii) Configuration of common parameters across multiple devices simultaneously;
 - (ix) Identification of inconsistent configuration between selected switches;
 - (x) Report generation of network statistics and other system information;
 - (xi) Capable of uploading configuration files and firmware updates to multiple switches at one time;
 - (xii) Capable of performing consecutive reboot to multiple switches automatically, one device at a time;
 - (xiii) Export of tabulated data in comma separated values (CSV) format;
 - (xiv) Open Productivity and Connectivity (OPC) interface to third-party monitoring;
 - (xv) Creation of user-defined parameters for third-party SNMP supported devices;
 - (xvi) User configurable dashboards; and
 - (xvii) Audit trail logs.
- (c) The NMS shall have the following features to assist in initial and subsequent network setup:
 - (i) All auto-discovered nodes shall be categorized in a hierarchical tree structure according to the physical or virtual location for easy management;
 - (ii) All graphical topology maps, port statistics, link status and alarms shall be able to be printed for documentation;
 - (iii) The network topology shall be able to be updated automatically, if the network is intentionally changed later;
 - (iv) New devices to the network, shall be detected automatically and a popup alarm shall be triggered; and
 - (v) Collation and display of network transmission performances. The network availability shall be calculated automatically.

G6.20.8 Edge Network Switches (Non-Process)

- (a) All non-process network switches, other than backbone switches, shall meet following requirements:
 - (i) Be of one make and one model family;
 - (ii) Be fully compatible with the overall network design. Any deviation from this shall be approved by the S.O.;
 - (iii) Be Layer 2 managed switches;
 - (iv) Have either 24 Vdc or 240 Vac power supply inputs as best suited for each specific situation;
 - (v) Have an operating temperature range of -40 °C to 60 °C (-40 °C to 70 °C in vented enclosure);
 - (vi) Have four, eight, or 16 10/100Base-T Ethernet ports with variants available having SFP downlinks;
 - (vii) Have two gigabit uplink ports being either SFP (100 Mbps and 1 Gbps) or RJ45 to suit the application;
 - (viii) Have model variants supporting Industrial and Enhanced Power over Ethernet (PoE);
 - (ix) Be conformally coated;
 - (x) Have IEC/EN 61000-4-5 2 kV/1 kV surge protection;
 - (xi) Have IEC/EN 61000-4-11 AC power Voltage Immunity;
 - (xii) Have IEC/EN 61000-4-29 Voltage Dips Immunity;
 - (xiii) Have IEC/EN 61000-6-2 EMC Immunity for Industrial Environments;
 - (xiv) Support Industrial Ethernet protocols including CIP, EtherNet/IP and PROFINETv2;
 - (xv) Support for IEEE 802.1x, Secure Shell (SSH) Protocol, Dynamic Host Configuration Protocol (DHCP) snooping, and storm control; and
 - (xvi) Support integration with the Network Monitoring System.

G6.20.9 FOBOTs

- (a) Fibre optic breakout trays ("FOBOT's") shall accommodate one 24 core fibre cable per 1 RU (rack unit) or one 48 core fibre cable per 2 RU (rack unit). FOBOT's and patch panels shall either be incorporated within other ICA panels, or shall be standalone panels, as shown in the Drawings.
- (b) FOBOT sizing shall match the cables being terminated so that there are no un-terminated ports on the FOBOT and no unterminated fibres on the cable.

END OF SECTION